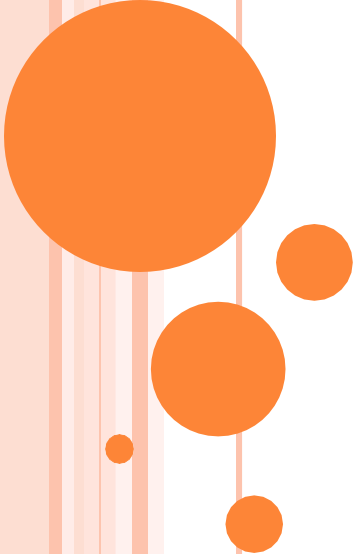




HS-401: Professional Values and Ethics

LECTURE NO 7: *Ethical Questions in the Design of Technology*

Engr. Tehseen Ahsan

**Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan**

Contents

1.1 Introduction

1.2 Ethical Issues During the Design Process

1.2.1 Problem analysis and formulation

1.2.2 Conceptual design

1.2.3 Simulation

1.2.4 Decision

1.2.5 Detail design

1.2.6 Prototype development and testing

1.3 Trade-offs and Value Conflicts

1.3.1 Cost-benefit analysis

1.3.2 Multiple criteria analysis

1.3.3 Thresholds

1.3.4 Reasoning

1.3.5 Value Sensitive Design

1.4 Regulatory Frameworks: Normal and Radical Design

1.1 Introduction (1/6)

Case High Speed Train Disaster in Germany



1.1 Introduction (2/6)

On June 3, 1998, the German high speed Intercity Express (ICE) train 884 “Wilhelm Conrad Röntgen” derailed at a speed of about 200 kilometers an hour and ran into a bridge that fell down on the train. The subsequent cars of the train jackknifed into the crashed cars in a zig-zag pattern. Overall, 101 people were killed and 88 severely injured. Investigations after the accident showed that the disaster was to a large extent due to a change in the wheel design of the train. Originally the train had been equipped with monobloc wheels, which are single casted. Such wheels decrease the rolling friction between the wheels and the rails, so increasing energy efficiency and lowering the costs of electricity consumption. However, such wheels may also decrease the comfort and may result in material stresses in the wheel. In some high speed trains, the comfort problem is solved by using a bogie (the framework carrying the wheels) with air suspension. The German ICE however was based on conventional bogies with steel springs. Once in service, it turned out that the wheel system caused severe resonance and vibration. This was considered a problem, especially in the dinner car where dinnerware “walked” over the tables. Engineers of the German railways then proposed to solve the problem with a new wheel design that had a rubber ring between the steel tire and the wheel body. This wheel design had been used in trams, but at significantly lower speeds. The new wheel design was not tested at high speed before it came into service, but was based on existing experience and materials theory. Nevertheless, after introduction the wheel solved the vibration problems and did not show any major problems until the fatal accident.

1.1 Introduction (3/6)

The new wheel design consisted of a steel wheel body surrounded by a 20 mm thick rubber damper and then a relatively thin steel tire. It was this tire that eventually failed and led to the disaster. This could have been caused by two mechanisms. One is metal fatigue due to the fact that the metal tire around the wheel is deformed a bit every rotation. The other mechanism is that the rubber damper extends due to heating up. Since rubber extends more than steel due to heat and since the rubber tire is locked in between the wheel body and the metal tire, this will result in a high pressure on the steel tire and, as a result, cracks may form from the inside in the metal tire. The first of these mechanisms was known at the moment that it was decided to change the wheel design, the second one not.

1.1 Introduction (4/6)

The wheels of the train were routinely checked every day. First, it was measured whether the wheels were still round enough or had become ellipsoids. The wheel out-of-roundness should not be larger than 0.6 mm. In practice, higher values were measured and allowed, probably because an out-of-round wheel was not seen as a safety issue but rather as an issue of comfort and wear. This might have been true for monobloc wheels, but it was certainly not true for the new wheel design. Second, the total wheel diameter should be at least 854 mm. A new wheel had a diameter of 920 mm. At the last check before the accident the wheels had a diameter of 862 mm. Later investigations of the Fraunhofer Institute in Darmstadt suggested that a norm of 890 mm would have been more appropriate given the new wheel design. Again, the norm was probably based on monobloc wheels, neglecting the peculiarities of the new wheel design. Third, the wheels were also inspected for cracks. Initially, advanced testing machines were used to inspect the wheels. These produced, however, a large number of false positives, that is, indications of a defect in the wheel while there actually was no problem. As result of these false positives, the use of the advanced inspection apparatus was discontinued and the wheels were only inspected visually with a flash light and audibly by sledging a hammer against the wheels. The latter method, which is often used for monobloc wheels, was probably inadequate for the new wheel design because the rubber damper will absorb the hammer stroke.

1.1 Introduction (5/6)

Three engineers, two from the German railways (Deutsche Bahn) and one from a supplier, who all had been involved in the certification of the wheels were tried and charged with manslaughter. After 53 days, the judge concluded that the three had not been grossly negligent. Three reasons were given for this. First, among experts there was not a principled objection against the new type of wheels. Second, the metal cracks that could cause an accident like this can develop rather quickly, so that adequate inspection procedures might not have prevented an accident like this. Third, it was not considered proven that the engineers had made gross mistakes in calculating the load on the wheels. Therefore the case was dismissed on the condition that each defendant paid a fee of 10,000 Euros (a possibility that is specific to German criminal law).

1.1 Introduction (6/6)

- The disaster with the German ICE train illustrates the importance of design to engineering. In this case, as in other cases, the design phase was crucial for the proper working of a technology and possible risks and other side effects. The example also shows the importance of activities that are closely related to design like testing, certification, and inspection. Second, the example shows that most design involves trade-offs. In wheel design for trains a trade-off is faced between energy (and cost) efficiency and comfort. Monobloc wheels are rather efficient because they cause little rolling resistance between the wheel and the track but they may negatively influence comfort(especially at high speeds).
- Finally, the example shows the importance and risks of innovative or radical design. The new wheel design was innovative in the sense that it had not been used for high speed trains. Although such innovative design may potentially solve important design problems, it may also create new risks.

1.2 Ethical Issues During the Design Process (1/4)

- The design process is a central area where ethical considerations regarding technology arise. The reason for this is that crucial decisions regarding technology are made in the design process. To an important extent the design of a technology determines how it will be produced and be used, what maintenance will be required, and how the product is to be scrapped.
- Designing can be described as an activity in which engineers translate certain functions or aims into a working product or system. A ferry can be conceived of as the translation of the function “transporting people from one side of the river to the other.” In most cases a function or social goal can be translated into a technical solution in several ways. If you want to achieve transport between two riverbanks, you can choose among a series of possible technical solutions, such as a bridge, a tunnel, a ferry, or a cable-lift. The solution chosen not only depends on the function to be realized, but on a series of additional design requirements, such as speed of transport, costs, building time, sustainability, and safety.

1.2 Ethical Issues During the Design Process (2/4)

- The *design process* is an iterative process that can be divided in six different stages, like:

1.2.1 Problem Analysis and Formulation

- Problem analysis stage: The stage of the design process in which the designer or the design team analyses and formulates the design problem, including the design requirements.
- Design requirements: Requirements that a good or acceptable design has to meet.

1.2.2 Conceptual Design

- Conceptual design stage: The stage in which the designer or the design team generates concept designs. The focus is on an integral approach to the design problem.
- Creativity: The virtue of being able to think out or invent new, often unexpected, options or ideas. Creativity is an important professional virtue for designers.

1.2 Ethical Issues During the Design Process (3/4)

1.2.3 Simulation

- Simulation stage: The stage of the design process in which the designer or the design team checks through calculations, tests, and simulations whether the concept designs meet the design requirements.

1.2.4 Decision

- Decision stage: The stage of the design process in which various concept designs are compared with each other and a choice is made for a design that has to be detailed.
- Design criteria: A kind of design requirements which are formulated in such a way that products meet them to a greater or lesser extent. Design criteria are often used to compare and choose between different concept designs.
- Trade off: Compromise between design criteria. For example, you trade off (compromise) a certain level of safety for a certain level of sustainability.

1.2 Ethical Issues During the Design Process (4/4)

1.2.5 *Detail design*

- Detail design stage: The stage in which a chosen design is elaborated on and detailed.

1.2.6 *Prototype development and testing*

- Test: The execution of a technology in circumstances set and controlled by the experimenter, and in which data are gathered systematically about how the technology functions in practice.

1.3 Trade-offs and Value Conflicts (1/5)

- As we have seen, in the decision stage, the design is assessed in terms of the design requirements. A choice has to be made between different designs. This choice is often far from simple, because the different designs usually score well on different criteria.
- Trade-off decisions are difficult decisions anyway but they may become particularly problematic if we face a *value conflict*.
- ***Value conflict***- A value conflict arises if: (1) a choice has to be made between at least two options for which at least two values are relevant as choice criteria, (2) at least two different values select at least two different options as best, and (3) the values do not trump(outclass) each other.

1.3 Trade-offs and Value Conflicts (2/5)

1.3.1 Cost-benefit analysis

- A method for comparing alternatives in which all the relevant advantages (benefits) and disadvantages (costs) of the options are expressed in monetary units and the overall monetary cost or benefit of each alternative is calculated.

1.3.2 Multiple criteria analysis

- Multiple criteria analysis is based on a comparison of different options with each other with respect to a number of criteria.

1.3.3 Threshold

- The minimal level of a(design) criterion or value that an alternative has to meet in order to be acceptable with respect to that criterion or value.

1.3 Trade-offs and Value Conflicts (3/5)

1.3.4 Reasoning

- An approach that does not share calculative approach, but which emphasizes judgment and reasoning about values. This approach aims at clarifying the values that underlie the conflicting design requirements, and consist of three steps: 1) identifying relevant values; 2) specifying the values; and 3) looking for common ground among values.

1.3.5 Value Sensitive Design

- One approach that takes into account the possibility of solving, or at least easing, value conflicts through engineering design is *Value Sensitive Design*.
- Value Sensitive Design is an approach that aims at integrating values of ethical importance in a systematic way in engineering design.

1.3 Trade-offs and Value Conflicts (4/5)

1.3.5 Value Sensitive Design(... Continued)

- Value Sensitive Design approach aims at integrating three kinds of investigations: conceptual, empirical, and technical:
 - *Empirical investigations* aim at understanding the contexts and experiences of the people affected by technological designs.
 - *Conceptual investigations* aim at clarifying the values at stake, and at making trade-offs between the various values.
 - *Technical investigations* analyse designs and their operational principles to assess how well they support particular values, and, conversely, to develop new innovative designs that meet particular morally relevant values.

1.3 Trade-offs and Value Conflicts (5/5)

1.3.6 A comparison of the different methods

<i>Method</i>	<i>How are the values weighted?</i>	<i>Main advantages</i>	<i>Main disadvantages</i>
Cost-benefit analysis	All values are expressed in monetary terms	• Options are made comparable	• Values are treated as commensurable • May be difficult to adequately express all relevant (moral) concerns in monetary terms
Multiple criteria analysis	Trade-offs between the different values	• Options are made comparable	• Values are treated as commensurable • Result depends on measurement scale
Thresholds	A threshold is set for each value.	• The selected alternatives meet the thresholds • No direct trade-off between the criteria	• Can thresholds be determined independently from each other? • Less achieved than possible
Reasoning	Values are related to each other and possibly traded off through reasoning and judgement.	• Might solve value conflict by reason and judgment	• Not all value conflicts can be solved in this way
Value Sensitive Design	Not applicable	• Can lead to alternatives that are clearly better than all of the present alternatives	• Does not solve the choice problem in many cases

1.4 Regulatory Frameworks: Normal and Radical Design (1/2)

- Engineering design often is redesign. In other words, something is designed that already exists in a comparable shape or based on a comparable working principle, such as a car, a bridge, a chemical plant, or a microchip. Experience with the product has often led to codes and standards for the design of that product and has also sometimes led to legislation from the government. Although some legislation and codes and standards apply to all engineers and technical products, many of the rules are more product-specific.
- *Regulatory framework*-The totality of (product-specific) rules that apply to the design and development of a product/technology.

1.4 Regulatory Frameworks: Normal and Radical Design (2/2)

- The existence of a framework meeting this set of conditions partly depends on the type of design process.
- **Normal design-** Design in which the normal configuration and working principle of the product remain the same.
- **Radical design-** The opposite of normal design. Design in which either the normal configuration or the working principle (or both) of an existing product is changed.

References: Textbooks

1. *“Ethics, Technology, and Engineering An Introduction”* by Ibo van de Poel and Lamber Royakkers, 1st Edition, Wiley-Blackwell, 2011.
2. *“Ethical and Social Issues in Information Age”*, by Joseph Migga Kizza, 6th Edition, Springer International Publishing AG 2017., Chapter 4.
3. *“Ethics in Computing, A concise module”*, by Joseph Migga Kizza, 6th Edition, Springer International Publishing Switzerland 2016.
4. *“Contemporary Issues in Ethics and Information Technology”*, by Robert A. Schultz, IRM Press 2006.
5. *“Professional Ethics and Human Values”*, by R.S. Naagarazan, New Age International Publishing 2006.
6. *“Engineering Ethics Concept and Cases”*, by Charles E.Harris, Jr.Michael S.Pritchard, Michael J.Rabins, 4th Edition, CENGAGE learning 2009.
7. *“Professional Values in Information Technology”*, by Frank Bott, The British Computer Society (BCS) 2005.
8. Related documents from open source, mainly internet.